Pre-analysis plan for a study of salience of study information and credibility assessment of key assumptions and inputs

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A note on this document. This pre-analysis plan was drafted with the assistance of Claude Code. It is unclear whether the survey will draw any responses at all. The plan was written for the optimistic case, in which enough people respond for a study to be conducted with the data, so that the analysis is fixed before any response is seen. **At this point, it is the opinion of the author that no human should spend much time reading this pre-analysis plan. It is only intended as a record.**

Status: the version registered on the AEA RCT Registry (trial 19997) on 2026-10-19, before the forms open. Structure follows the minimal elements of a clear hypothesis: population, interventions, arms, outcomes, hypotheses (contrasts, tests, alternative, estimation, standard errors), and heterogeneity stated separately and in full.

1. Question

The Proposition 40 survey asks respondents which of two proposed values of each disputed input they find more credible. The comparison explorer shows the same inputs, which study uses which value, and what each value does to the estimate. Most respondents will have seen that comparison on the site before they open the survey. This study asks whether restating it, with each value labelled by its study, at the moment respondents choose the input values changes the answers.

2. Population and sample

- **Population.** Everyone who opens one of the project's survey links and submits the form during the collection window (section 9). No one is screened out. Self-described experts and respondents without prior exposure are subgroups (section 6), not the sample.
- **Unit of randomization.** The respondent, drawn at the click on a survey link (section 4.2).
- **Unit of analysis.** One submitted response.
- **Sample size.** Not controlled: it is whoever responds in the window. The number of responses depends on whether the launch of the OPA gains traction on social media, and it is likely that there will be none or very few. A reasonable range is 0 to 200. The staged rule of section 7 decides which analyses run once the realized number is known, and section 7 gives the detectable effects for a range of sizes.

3. Interventions

Two manipulations, both delivered only in the treatment forms. Nothing is delivered to the control group: it is a pure control.

- **I1, the comparison page.** One page, "How the two estimates differ", shown after the opening questions and before the input questions. It opens with "Two academic studies" and gives each study on its own bullet with its month and its estimate (Boll, Saez and Zucman, BSZ; Jaros and Rauh, JR). Neither bullet names a venue: both are NBER papers, and a venue on one side only would be a prestige cue inside the treatment. It then gives the gap between them and the explorer's common measure. It then splits the gap into the four terms of the comparison explorer's equation, largest first. Each term has a bar, its amount in billions and its share of the gap, and, for the first three, the values each study uses. A sentence names the two largest terms and their combined share. A two-sentence note says why the lost income tax is one term (below) and that each amount depends on the order in which the inputs are changed. The full text is in Appendix A.
- **I2, the study labels.** In the input questions, each choice is labelled with its study ("BSZ:" or "JR:") instead of "A." or "B.". The choice text and order are otherwise unchanged.

The four terms, as generated at the launch commit (section 9; gap $145.8 billion):[1]

Term on the page	Amount	Share of the gap
The income tax lost when billionaires leave: the share lost, how long it is counted, and billionaires' yearly income tax, together	-$81.3 billion	56%
Discounting the revenue for court risk	-$39.4 billion	27%
Billionaires who moved out in time	-$29.9 billion	21%
Other inputs together: the wealth on the list, avoidance, and three smaller inputs	+$4.9 billion	offsets 3%

The shares add to 101% because of rounding.

The page does not give one amount per survey question. The lost income tax is a product of the share lost, how long it is counted and the yearly amount. A split among those three factors would depend on the order in which they are changed, so it would misstate what any one of them does. The page shows them as one term and says why. As a result, the terms do not match the survey questions one to one. The first term covers three whole questions: the share lost, how long it is counted, and the yearly income tax. The question on billionaires who moved out in time spans two terms. The JR value (seven named moves) is the third term. The BSZ value (an allowance for unnamed movers, half of the BSZ avoidance allowance) sits in the fourth. The two optional questions, on avoidance and on the wealth the tax applies to, also sit inside the fourth term. Only the court question matches one term. The small difference in discount rates is inside the first term and is not named, since the horizon question no longer asks about it.

[1] Each amount is the sum of the explorer's walk steps in that term (`comparison/export/r/dial_path.csv`). The walk moves the inputs one after another from the BSZ values to the JR values. The values come from `comparison/export/r/inputs.csv`. `comparison/survey/mock_forms.py` and `gas_build.py` generate the page from those files, so no number is typed. They also check that every walk step belongs to one term and that the terms add up to the gap.

4. Arms and assignment

4.1 Arms

Arm	I1	I2
Control (T = 0)	no	no
Treatment (T = 1)	yes	yes

I1 and I2 are delivered together. The design measures their joint effect and cannot separate them.

4.2 Assignment

Google Forms cannot randomize inside a form, so the draw happens at the click. Each survey link opens its entry point's control form or treatment form with probability 1/2, drawn in the browser (`Math.random() < 0.5`). The respondent sees nothing of either form before the draw. Draws are independent, and each entry point is a stratum.

4.3 Entry points and forms

Four forms: two entry points, two arms each. The form a response lands in records its entry point and its arm. Respondents see neither label. Appendix B reproduces the control and treatment forms.

Entry point	Who arrives there	Control	Treatment
`explorer`	the survey button lower on the explorer page, and the overview page's survey button	form 3	form 4
`share`	the share link (`comparison/site/survey/`): copied from the site, sent by email, or opened from the public posts through tinyurl.com/prop40-survey	form 5	form 6

Forms 1 and 2 belonged to a third entry point, `landing`: a panel shown to every visitor of the explorer before the results, inviting them to take the survey first. The panel was removed on 2026-10-17, before launch, because it stood between visitors and the explorer and could deter them from using it. Forms 1 and 2 were built but are never opened.

The OPA pages point readers to the comparison explorer several times, and the explorer shows the two competing estimates side by side. Every respondent who arrives through the `explorer` entry point has therefore seen the comparison before opening the survey. The `share` link opens a form directly, so its respondents are a mix: some have seen the estimates and some have not. For this reason the experiment mainly tests the salience of that information when respondents choose the input values, not their first exposure to it.

4.4 Page order (all forms)

1. Description, then two pre-treatment questions and the optional email:

- "Do you consider yourself an expert in tax or fiscal policy?" (Yes / Somewhat / No; moved here from the end of the current survey)
- "Before opening this survey, had you looked at news sources, web pages or academic studies discussing estimates of Proposition 40, the Billionaire Wealth Tax Proposal in California?" (Yes / No / Not sure; new)
- "Your email (optional)" (moved here from the closing page on 2026-10-18, so it is asked before treatment; nothing is verified)

2. I1 (treatment forms only).

3. The five required input questions, in three groups under plain section headings, in the order of the treatment page's first three terms (`GROUPS`, `regroup()` and `split_bundle()` in `mock_forms.py`):

   - "1. The income tax lost when billionaires leave": "The share of billionaires' income tax lost when people leave: which is more credible?", then how long the lost income tax is counted, then billionaires' yearly income tax. The three are ordered by each input's effect on the estimate averaged over all orders of the dials, largest first.
   - "2. Discounting the revenue for court risk": the court question.
   - "3. Billionaires who moved out in time": "Billionaires who moved out of California before the 1 January 2026 residency date: which is more credible?" It spans two of the page's terms (section 3).

   Then the opt-in to the five smaller inputs, whose page ("Five smaller inputs (optional)") follows: wealth that escapes the tax through avoidance and evasion, the wealth the tax applies to, non-citizen residents, the real-estate deduction, and the extra income tax from assets sold to pay the tax.

   The current survey bundles departures, avoidance and the share lost into one question. In all the forms that question becomes three: the share lost and moved out in time (required), and avoidance (optional), so both choices of each answer the same thing. The wealth question moves from the required questions to the optional page. I2 applies to all ten input questions in the treatment forms.

4. The closing page, "Anything else (optional)": missing inputs and support for Proposition 40. Every question on it is optional.

The question order and the three headings are the same in both arms, so they are not part of the treatment. They differ from the current survey's order. All the forms use one fixed choice order within each question, that of the current version 1. The simulated pilot (section 9) showed no sign that the first-listed value gains from its place: in the control forms the second-listed value won the share-lost question (21 to 9) and the court question (18 to 4). The horizon question drops the discount rate (author, 2026-10-16): "How long the lost income tax is counted: which is more credible?", with the choices "For ever: billionaires who leave are assumed not to return." and "5 years, about the payment period, treating the departures as temporary." ("departures" replaced "responses" on 2026-10-17, a word no respondent had been given a meaning for). The current survey's "Which field best describes your work?" is dropped (author, 2026-10-16).

Wording of the values (2026-10-17, after the simulated pilot; approved by the author the same day). In every input question the two values are the same kind of claim. Each gives

a reason, or neither does. Neither value argues against the other. The texts are generated by `mock_forms.py` (`SHARE`, `DEPARTED`, `AVOID`, `INCOME`, `COURT`, `WEALTH`, `NONCIT`, `REAL`, `ASSET`), with every number read from the exports. What changed, and why:

- The share lost now states both values on the same base, "of all the income tax that billionaires pay" (`comparison/DISPUTES.md`, row 6): 45%, more than the 29% of wealth reported to have left, against 5%, the same as the 5% of wealth assumed to leave. Each comes with its analysis's own reason. The old pair ("45% of all billionaires' income tax, whoever leaves" against "5%, only the income tax of those who leave") was read in three ways in the pilot: as a count of leavers, as shares of two different bases, and as losing the tax of those who stay. A respondent who accepts one analysis's count and the other's rule can now answer.
- The court question now asks whether to bring a court-risk discount into the estimate at all, "Discounting the revenue for the chance that courts strike the tax down": no discount, the tax assessed at face value, against the wealth tax collected multiplied by the chance the tax survives, put at 50%. The discount applies to the wealth tax collected, not to the whole estimate (JR chapter, PDF p.33; the comparison applies q to the wealth tax only). It drops "guesstimated", which all ten interviewees flagged, and neither value sounds careless. The question on billionaires who moved in time gives each value its own standard for counting a move. It no longer contains a rebuttal of the other value, or the third value (the "aggressive scenario") that interviewees read against both sides.
- The yearly income tax gives each value one assumption with its source, in the same words: the JR value names its two counts of top earners (212 and 500, read from the export's source note), the BSZ value the study of top wealth holders. The old JR value carried a range and its two bounds and the BSZ value one line ("halved"). The pilot split 30 to 3 on it, and six of ten interviewees named the difference in length.
- Four of the five optional questions give both values the same form ("Taxed" and "Not taxed", "Counted" and "Not counted"). In the pilot the bare JR values ("None.", "Not counted.") drew 0 or 1 choices each. The avoidance question keeps its pilot text ("5%: a small allowance argued from enforcement." against "None."), the author's choice. The BSZ wealth total now says how much of it is the non-US citizens asked about next.
- The description no longer promises "the reason given for each" value, says "most of these are empirical quantities", and says 5 to 10 minutes (careful readers took a median of 14 in the pilot). It ends with a sentence of motivation: the answers help get closer to agreeing on common facts for policy discussions. The opt-in says that the next five questions explain up to a stated share of the difference between the analyses (the net of the five dial-path steps over the gap, the explorer's "other" term) and make the survey a stated share longer (the optional section's words over the main part's), and that answering them is optional. The email question is "Your email (optional)": nothing is verified. After submitting, every form shows the same confirmation: thanks, the motivation sentence, and two short links, tinyurl.com/prop40-explorer and tinyurl.com/prop40-opa.

5. Outcomes

5.1 Coding of one answer

For respondent i and required question k ($k = 1$ to 5, in the order the forms ask them: the share of billionaires' income tax lost; how long the lost income tax is counted; billionaires' yearly income

tax; the chance the tax survives in court; billionaires who moved out in time):

- $y_{ik} = +1$ if the answer is the BSZ value,
- $y_{ik} = -1$ if the answer is the JR value,
- $y_{ik} = 0$ if the answer is "About equally credible" or "Unsure".

The two neutral answers mean different things. "Unsure" is a don't-know: the respondent cannot judge. "About equally credible" is a judged tie: the respondent has judged and neither value wins. Both are offered because dropping an explicit don't-know would push respondents without a view onto the middle answer. Both score 0 in the lean index, and S1 and S2 report them separately.

In the treatment forms the label gives the study. In the control forms the private Key tab of the responses sheet gives which of A and B is the BSZ value in version 1's order. All five questions are required, so no value is missing. The avoidance and wealth questions are now optional (section 4.4), so they are not among the five. In the share-lost question both values are shares of the same total, the income tax billionaires pay, each shown with its analysis's reason.

5.2 Primary outcome

The lean index, $\text{Lean}_i = (y_{i1} + y_{i2} + y_{i3} + y_{i4} + y_{i5})/5$. It runs from -1 (the JR value on all five) to $+1$ (the BSZ value on all five). The primary outcome is the index, not any one of its inputs.

5.3 Secondary outcomes

- S1. The share of the five questions answered "Unsure".
- S2. The share answered "About equally credible".
- S3. Each of the five y_{ik}, reported as three indicators (BSZ value, JR value, neutral answer).

In the simulated pilot, "About equally credible" was 2% of the required answers and "Unsure" 45%. The reasons given with "Unsure" on the horizon and court questions often said that neither value was right ("100% is too high, but 50% is a guess"). S1 therefore mixes not knowing with rejecting both values. The author keeps the two answers as they are (2026-10-17); the reasons given with "Unsure" are reported alongside S1.

The five optional smaller-input questions, avoidance and the wealth base among them, are excluded from every registered outcome, because only respondents who opt in answer them. Their answers are reported descriptively.

6. Hypotheses and estimation

6.1 Main hypothesis (primary)

H1. $E[\text{Lean} \mid T = 1] - E[\text{Lean} \mid T = 0] = 0$, against the two-sided alternative that it is not 0.

Estimated by OLS with entry-point fixed effects (the strata):

$$\text{Lean}_i = \alpha + \beta\, T_i + \gamma\, E_{\text{share},i} + \varepsilon_i$$

H1 is $\beta = 0$. Standard errors are heteroskedasticity-robust (HC2). The test is at the 5% level. The robustness specification adds the two pre-treatment answers as indicators.

Secondary outcomes S1 to S3 are estimated with the same equation. The five S3 outcomes are adjusted with the Romano-Wolf step-down procedure within each answer type (section 6.3).

6.2 Heterogeneity (all secondary)

Every heterogeneity test below is secondary. None changes the conclusion on H1.

The plan declares D = 3 candidate dimensions, all binary and recorded before treatment: entry point (H2), prior exposure (H3) and expertise (H4). Whether a respondent gives an email is asked before treatment, on page 1, but it is not registered as a dimension, so it stays descriptive and is not one of the three. No other dimension is used unless it is registered before the data are seen, and adding one enlarges every family below.

H2 to H4 are the one-dimension analyses. With H5 below, they form one family of four tests ($\binom{3}{1} = 3$ and H5), adjusted together with the Romano-Wolf procedure. Each is one interaction coefficient, estimated in the H1 equation with the subgroup indicator and its interaction with T added:

$$\text{Lean}_i = \alpha + \beta\, T_i + \delta\, (T_i \times G_i) + \lambda\, G_i + \gamma\, E_{\text{share},i} + \varepsilon_i$$

The test is $\delta = 0$ against $\delta \neq 0$, with HC2 standard errors.

- **H2, entry point.** $G_i = 1$ if the response came through the `explorer` button, 0 if through the `share` link. Prediction: the effect is smaller in absolute value for `explorer`, whose respondents have all seen the comparison on the page. Some `share` respondents have not. The entry point is not randomized: differences in levels between entry points describe who arrives where. Within each entry point, T is randomized. For H2, G is the entry-point indicator itself, so λG_i is dropped.
- **H3, prior exposure.** $G_i = 1$ if the answer to the prior-exposure question is "No", 0 if "Yes" or "Not sure". Prediction: the effect is larger in absolute value for G = 1.
- **H4, expertise.** $G_i = 1$ if the answer to the expertise question is "Yes" or "Somewhat", 0 if "No". No direction is predicted.
- **H5, experts without prior exposure.** The subgroup is the respondents who answer "Yes" or "Somewhat" to the expertise question and "No" to the prior-exposure question. H5 is that the treatment effect on the lean index inside this subgroup is 0, against the two-sided alternative. It is estimated with the H1 equation on the subgroup's responses only, with HC2 standard errors. These respondents know the field but had not seen estimates of Proposition 40, so for them the treatment page is closest to a first exposure. H5 is tested together with H2 to H4, in one Romano-Wolf family of four tests, and is run only once N reaches that family's threshold (section 7).

Analyses that cross two or three dimensions are staged secondary analyses (section 7). Crossing k dimensions gives 2^k cells. The k-way difference of effects is the coefficient on $T \times G_1 \times \cdots \times G_k$ in the saturated version of the H1 equation, with HC2 standard errors. Each order is its own Romano-Wolf family over every combination of the three dimensions: $\binom{3}{2} = 3$ two-way differences, and one three-way difference. The effects inside the cells form their own families: 6 for one dimension (3 dimensions x 2 halves), 12 for two (3 pairs x 4 cells), and 8 for three.

Both subgroup questions are asked before the treatment page, so treatment cannot change them. Support for Proposition 40 stays on the closing page, after treatment, and is not used for any registered test (decided 2026-10-17: asking it first would prime both arms). It is optional, so the share who answer it is reported by arm alongside its descriptive results.

6.3 Multiple testing

Every family of secondary tests (the five questions of S3, the one-dimension interactions with H5, the two-way and three-way differences, the effects inside cells) is adjusted with the Romano-Wolf step-down procedure (Romano and Wolf 2005; the algorithm of Clarke, Romano and Wolf 2020). The procedure controls the family-wise error rate over the bootstrap distribution of the largest studentized statistic in the family, so correlated tests cost less than under Holm or Bonferroni, and the five questions and the index are correlated. The statistics are the HC2 t-statistics of the registered equations. The bootstrap resamples respondents with replacement within each entry-point stratum, 2,000 draws, seed fixed before the data are seen. The power appendix states the thresholds under this procedure and the Bonferroni bound they sit below.

7. Power and staged analyses

The number of responses is unknown: the expected range is 0 to 200 (section 2). The plan therefore has one main hypothesis with one outcome (H1, the lean index), tested whatever the number of responses. Further analyses are run only once the realized number of responses N reaches the threshold at which each can detect a standardized effect of 0.5 SD with 80% power. The thresholds, their formulas and simulations of the registered estimator are in the power appendix, `comparison/survey/power/power.pdf` (built from `power.tex` and `sim.py`, which generates every number in it).

Summary (the appendix's Table 1): responses needed for 80% power against an effect of 0.3 SD, the two entry points at 60% (`explorer`) and 40% (`share`) and other halves equal, the Romano-Wolf step-down within each family. "RW" is the simulated Romano-Wolf threshold; "Bound" is the closed-form Bonferroni bound it sits below. A dagger marks a family whose tests are nearly independent, where the simulation found no threshold below the bound and the bound is used. For the five questions, column A is 0.3 SD on a question and column B is the smaller question-level effect (about 0.18 SD) that goes with 0.3 SD on the index. For heterogeneity, column A is the effect inside each cell and column B is the highest-order difference of effects. Each family covers every combination of the three dimensions.

	Analysis	A: tests	A: RW	A: bound	B: tests	B: RW	B: bound
1	Main hypothesis H1, the lean index	1	349				
2	H1 by outcome, each of the five questions	5	440	520	5	1,110	1,410
3	Heterogeneity, one dimension at a time (3 combinations, 2 cells each)	6	920	1,346	4	1,983 (dagger)	1,983

	Analysis	A: tests	A: RW	A: bound	B: tests	B: RW	B: bound
4	Heterogeneity, two dimensions at a time (3 combinations, 4 cells each)	12	2,010	3,054	3	7,090	7,445
5	Heterogeneity, three dimensions at a time (1 combination, 8 cells)	8	3,850	5,684	1	23,256	

Staged rule (the appendix's Table 4), at the registered target of 0.5 SD with the same shares. The Romano-Wolf threshold governs; the bound is given for reference.

Tier	Analysis	Tests	Responses needed (RW)	Bound
0	H1, the lean index	1	always run; detects 0.5 SD from 126	
1	Each of the five questions	5	150 (about 400 for an effect matching 0.5 SD on the index)	187 (497)
2a	One dimension at a time: the effect inside each half (expertise, prior exposure, entry point; email descriptively)	6	330 (540 if one half holds a quarter)	485 (776)
2b	The difference between the halves (H2 to H4) and H5, experts without prior exposure	4	714 (dagger) (950 if one half holds a quarter)	714 (952)
3a	Two dimensions at a time: the effect inside each of four cells	12	740	1,100
3b	Two dimensions at a time: the two-way difference	3	2,681 (dagger)	2,681
4a	Three dimensions: the effect inside each of eight cells	8	1,380	2,047
4b	Three dimensions: the three-way difference (one test)	1	8,373	
5	Each question inside each half, one dimension at a time	30	390 (about 1,070 for an effect matching 0.5 SD on the index)	509 (1,351)

The 2b threshold assumes the H5 subgroup holds a quarter of the responses. If it holds half, H5 alone would need 357, and H2 to H4 set the threshold. Explorer respondents have all seen the comparison, so the subgroup will sit mostly in the `share` stratum; the rule is applied at the realized shares.

Heterogeneity is unlocked one order at a time. Two-dimension analyses are run only if the one-dimension analyses of the same kind are run, and three-dimension analyses only if the two-dimension ones are. Each added dimension multiplies the responses a difference of effects needs by about four, because the cells halve in size and the difference adds cells.

The rule uses only N and the realized shares of pre-treatment subgroups at the close, so it does not depend on the results. An analysis below its threshold is reported as "not run: below its threshold".

The table's numbers come from the appendix (`out/summary.tex` and `out/thresholds.tex`); if the appendix is regenerated, it governs. The appendix's simulations use 199 bootstrap draws inside each simulated sample, fewer than the registered 2,000, so its thresholds lean high.

In the lean index's own units, with the simulated index SD of about 0.5 (an assumption, not a measurement), 0.5 SD is a change of 0.25 in the mean index. That is, for example, 125 one-step answer moves per 100 treated respondents (a step is neutral to a study's value, or the other value to neutral), or a quarter of treated respondents moving all five answers one step. 0.4 SD is 0.20 (100 moves per 100, or 20% moving all five answers). 0.3 SD is 0.15 (75 moves per 100, or 15%). 0.2 SD is 0.10 (50 moves per 100, or 10%). 0.1 SD is 0.05 (25 moves per 100, or 5%). The appendix's Table 3 and Figure 1 give these, with the responses H1 needs at each size.

8. Checks, attrition and exclusions

- **Sample.** Every response submitted to the four open forms (3 to 6) in the window. Excluded: test responses made before launch (deleted from the sheet), and responses to the two current forms, which predate the experiment and are reported on their own.
- **Attrition.** Google Forms records only submitted responses. If the comparison page makes people quit, the treated respondents would differ from the controls. Checks, reported whatever the result:

    1. Within each entry point, the number of submissions per arm against an even split (exact binomial test).
    2. The two pre-treatment answers compared across arms (chi-squared tests).
    3. A count of draws. Each click on a survey link sends one request, at the draw, carrying only the entry point and the arm (no identifier, no cookie). The registered attrition check is submissions divided by draws, by arm and entry point. The binomial test above is the fallback if the count fails. With 100 submissions it detects only a split of about 60 to 40, and a respondent who quits on the comparison page leaves no other trace.

- **Duplicates.** Of responses with the same email, only the first is kept. Anonymous duplicates cannot be detected. One person can land in both arms, which shrinks any measured effect.
- **Bulk and automated submissions.** Google Forms does not screen out automated submissions. A bot that posts to one form's address skips the draw at the click, so it fills one arm only. That would bias the estimate, not only add noise. The rule is fixed before launch. Three or more submissions to the same form within ten minutes, with identical answers to every choice question, count as one submission, and only the first is kept. The number removed per form is reported. H1 is also reported with no response removed. A burst in one arm also shows in the per-arm count check above.
- **Respondents who give an email.** Their answers are also reported separately, as the survey promises. The registered analysis uses all responses.
- **Links that skip the draw.** The share page sends the browser to the form's own address, which a respondent can copy and forward. Everyone who opens a forwarded form address lands in one arm. The share page's fallback for browsers without JavaScript also opens one form. Neither group can be identified in the data. The Copy link button copies the share page's address, not a form's address (the public posts no longer ask readers to share it, section 11), and the fallback opens the share pair's control form. A form with many submissions and few draws is the sign of a forwarded address.

9. Timeline, registration and priors

- **Registration.** On the AEA RCT Registry, before the four forms open.
- **Launch.** The forms open together with the site change. The launch commit is recorded here.
- **Close.** 4 November 2026. Responses submitted after it are not part of this analysis. Responses submitted after the vote of 3 November 2026 and before the close count. The time of submission is recorded, and an indicator for "after the vote" is a covariate in the robustness specification of H1, reported alongside the main estimate.
- **Pretest.** On 2026-10-17, 100 simulated respondents (language-model agents given personas, entry paths and arms) took mock versions of the six forms, and ten of them were interviewed (`feedback/008-2026-10-17-rct-pilot/`). The pretest tested wording and comprehension only. Its answers are not data for this study. The question texts were revised after it (section 4.4). A second pretest the same day, with 100 new simulated respondents and no interviews, ran the revised forms through this plan's analysis (`feedback/009-2026-10-17-rct-pilot-2/`); its answers are not data either.
- **No interim analysis.** Before the close, only response counts per form are looked at.
- **Priors.** No forecasts of the effect are collected.
- **Deviations.** Any change after registration is logged at the end of this file with its date and reason.

How to launch.

1. Run the Apps Script that `comparison/survey/gas_build.py` writes. It builds the six forms (forms 1 and 2, for the dropped `landing` entry point, are never opened), named "exp: Prop 40 survey (,)". Each form takes about 110 seconds and a run stops at six minutes, so a run builds three forms; before running again, move the half-built fourth form to the trash (the builder skips any form whose name exists), then run again.
2. Paste each form's respondent link into `comparison/data/survey-links.csv`, in the row of its entry point and arm (`python3 comparison/survey/fill_links.py LOG.txt` does it from the execution log). Delete the comment line about placeholders. 2b. The thank-you screen after Submit carries the two short links (tinyurl.com/prop40-explorer and tinyurl.com/prop40-opa); `fixEnding()` sets it in forms built before 2026-10-17 afternoon. No link sits before Submit: a control respondent could open it, go Back and change answers.
3. Deploy `comparison/survey/draws.gs` as a web app (execute as me, anyone can access), as its header says. Paste its URL into `comparison/data/survey-config.csv` as `draws_url`.
4. Rebuild with `Rscript comparison/run.R`, then run `Rscript comparison/tests/testthat.R`.
5. Commit, and record the commit under "Launch" above.
6. Register on the AEA RCT Registry. Then push: the push puts the four forms' links on the site.

10. Limits

- **Who answers.** The results describe people who take the survey, not experts in general. The randomization makes the two arms comparable, not representative.
- **Prior exposure.** Every `explorer` respondent, and many `share` respondents, will have seen the comparison before the survey. The treatment therefore restates information most

respondents already have, which likely shrinks the effect. H2 and H3 are there to show it.

- **A bundled treatment.** I1 adds how much of the gap each of four groups of inputs explains. I2 adds which study holds each value. Naming the studies may move answers through views of the authors or of the tax, not only through the numbers. H1 measures the two together.
- **Terms and questions differ.** The page's four terms do not match the five questions one to one (section 3). Only for the court question does the page give that input's own amount. A question-level effect (S3) therefore cannot be read as a response to that question's amount on the page.
- **Neutrality.** The control forms keep the survey's rule of naming neither analysis. The treatment forms name both, by design.
- **The sides can be told apart without labels.** In the simulated pretest, the persona's view of wealth taxes predicted the lean index as strongly in the control forms as in the treatment forms (a gap of 0.68 between the personas who favour and oppose them, against 0.71), because a value's content shows which side it helps. Respondents from the explorer have also seen each value's study on the dials. The labels may therefore add less than their presence suggests, which shrinks the effect that H1 can find.
- **What the entry texts promised.** The drafts of the public post that carries the share link said the survey "names neither analysis", which is false for the treatment forms; in the pretest, ten of the eighteen simulated respondents who came by that link to a treatment form noticed the contradiction. The drafts were edited on 2026-10-17, before posting: no promise about naming, "5 to 10 minutes", and a request to pass on the project link rather than the form's address. That request was removed on 2026-10-18 (section 11).

11. Decisions taken

All of the open decisions were settled on 2026-10-17: the two current forms close when the new forms open; the overview's survey button sends to the `explorer` pair; the value texts revised after the pretest (section 4.4); the treatment page (Appendix A); the bulk-submission rule, the draw count and the links that skip the draw (section 8); the former secondary outcome S4 (the mean absolute lean index), dropped by the author on 2026-10-18; the close date and the post-vote responses (section 9); the Romano-Wolf adjustment (section 6.3); the two neutral answers, kept as they are (section 5.1); the public post, edited before posting (section 10); and the support question, kept after treatment (section 6.2).

Later on 2026-10-17, before launch, the author made four more decisions: the `landing` entry point and its panel are dropped, so forms 1 and 2 are never opened and the experiment is framed as a test of salience (sections 1 and 4.3); the expected number of responses is 0 to 200 (section 2); the expert respondents without prior exposure form a registered subgroup (section 6.2); and the share link keeps opening a form, so that some respondents may not have seen the estimates.

On 2026-10-18, before launch, the author moved the email question from the closing page to page 1, after the two pre-treatment questions (section 4.4). It stays descriptive (section 5). The form's description no longer says that the email comes near the end, and the opt-in's stated extra length fell from 60% to 55%, because the main part now includes the email question.

Also on 2026-10-18, before launch, the author removed from the public posts (LinkedIn and Bluesky) the request to pass on the project link rather than the form's address (sections 8 and 10). A respondent who forwards a form's address sends the people who open it to one arm, with no draw. The sign of this in the data is unchanged: a form with many submissions and few draws (section

8).

Appendix A. Treatment page text

Generated by `python3 comparison/survey/gas_build.py ITEMS.json SOURCE_FORM_ID` (the Apps Script that builds the forms) and previewed by `python3 comparison/survey/mock_forms.py ITEMS.json OUT_DIR`. The text comes from `STUDIES_LEAD`, `STUDIES`, `INTRO`, `TERMS`, `NOTE` and `CLOSE` in `mock_forms.py`. The page has a title, a lead sentence ("Two academic studies estimate what Proposition 40 is worth to California:") with one bullet per study, its month and its estimate, two paragraphs (the gap and the common measure; the two largest terms and their share), the four terms of section 3 with the BSZ and JR values of the first three, a two-sentence note (why the lost income tax is one term, and that each amount depends on the order in which the inputs are changed), and one closing sentence. The full text, as generated at the launch commit, follows.

How the two estimates differ

Two academic studies estimate what Proposition 40 is worth to California:

- Boll, Saez and Zucman (BSZ, August 2026): +$106.8 billion.
- Jaros and Rauh (JR, September 2026): -$39.0 billion.

The gap is $145.8 billion. Both are on one measure: the wealth tax collected, plus the extra income tax from assets sold to pay it, minus the income tax lost when people leave.

Two parts explain 83% of the gap: the income tax lost when billionaires leave, and the chance the tax survives in court. Below, how much of the gap each part explains, with each study's values for the first three.

1. **The income tax lost when billionaires leave: the share lost, how long it is counted, and billionaires' yearly income tax, together**: $81.3 billion, 56% of the gap. BSZ: 5% of all the income tax that billionaires pay is lost; counted for 5 years; $3.06 billion a year. JR: 45% of all the income tax that billionaires pay is lost; counted for ever; $4.55 billion a year.
2. **Discounting the revenue for court risk**: $39.4 billion, 27% of the gap. BSZ: no discount. JR: a 50% discount.
3. **Billionaires who moved out in time**: $29.9 billion, 21% of the gap. BSZ: no billionaire counted by name (its allowance for 5% of the wealth moving away in time is in part 4). JR: seven billionaires counted on public reports of their moves.
4. **Other inputs together: the wealth on the list, avoidance, and three smaller inputs**: +$4.9 billion, offsetting 3% of the gap.

The income tax lost is shown as one amount because its three inputs multiply each other. For the same reason, each amount depends on the order in which the inputs are changed.

The questions that follow ask which study's value of each input you find more credible.

Appendix B. The two forms

The pages that follow reproduce the control form (form 3) and the treatment form (form 4) as respondents meet them, page by page, with the confirmation message shown after Submit. Each copy is printed from the form's own definition as Google Forms served it in preview on 18 October 2026 (`comparison/survey/plan/forms/control.json` and `treatment.json`, printed by `print_forms.py` in the same folder). Forms 5 and 6, the `share` entry point's forms, carry the same text as forms 3 and 4; only Google's internal item ids differ.

Appendix B.1. The control form (forms 3 and 5)

Printed from the form's own definition as Google Forms served it in preview on 18 October 2026. Each page of the form starts at a purple rule. Respondents see the text exactly as below, one page at a time.

PAGE 1 OF 4

What are the actual quantities? Proposition 40, input by input

Proposition 40 would tax Californians worth $1 billion or more once, at 5% of their net worth. Estimates of what it is worth to the state depend on a handful of inputs. Each question shows one input and two values that have been proposed for it. Tell us which value you find more credible. Your answers will help us get closer to agreeing on common facts to inform important policy discussions.

Most of these are empirical quantities, and none should depend on whether one supports or opposes a wealth tax.

It takes 5 to 10 minutes. Answers are anonymous unless you leave an email.

* Indicates required question

Do you consider yourself an expert in tax or fiscal policy? *

- ○ Yes
- ○ Somewhat
- ○ No

Before opening this survey, had you looked at news sources, web pages or academic studies discussing estimates of Proposition 40, the Billionaire Wealth Tax Proposal in California? *

- ○ Yes
- ○ No
- ○ Not sure

Your email (optional)

If you leave one, your answers are also reported in a separate analysis of the respondents who gave an email.

Your answer

The inputs

For each input, pick the value you find more credible.

1. The income tax lost when billionaires leave

The share of billionaires' income tax lost when people leave: which is more credible? *

- ○ A. 45% of all the income tax that billionaires pay: more than the 29% of their wealth reported to have left, because income tax need not leave in step with wealth and some moves go unreported.
- ○ B. 5% of all the income tax that billionaires pay: the same as the 5% of their wealth assumed to leave, because the largest of the billionaires reported to have left pay little income tax for their wealth.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

How long the lost income tax is counted: which is more credible? *

- ○ A. For ever: billionaires who leave are assumed not to return.
- ○ B. 5 years, about the payment period, treating the departures as temporary.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

Billionaires' yearly California income tax: which is more credible? *

- ○ A. $4.55 billion a year: from California tax data, taking the billionaires to be among the state's 212 to 500 highest earners.
- ○ B. $3.06 billion a year: from California tax data, taking the billionaires to have about half the income of the state's highest earners, as a study of top wealth holders finds.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

2. Discounting the revenue for court risk

Discounting the revenue for the chance that courts strike the tax down: which is more credible? *

- ○ A. No discount: the tax is assessed at face value, as written.
- ○ B. A discount: the wealth tax collected is multiplied by the chance the tax survives in court, put at 50%.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

3. Billionaires who moved out in time

Billionaires who moved out of California before the 1 January 2026 residency date: which is more credible? *

- ○ A. About 29% of the wealth: seven billionaires (Page, Thiel, Hankey, Kalanick, Brin, Spielberg and Sacks) are counted as having moved in time, on public reports of their moves.
- ○ B. 5% of the wealth: an allowance for billionaires who moved in time, with no one counted by name without evidence of a change of legal residence before the date.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

The next 5 questions in total explain up to 3% of the difference between the two analyses. Answering them makes the survey on average 55% longer. Answering them is optional. *

- ○ Yes, show me the five (goes to the page “Five smaller inputs (optional)”)
- ○ No, skip them (goes to the page “Anything else (optional)”)

Five smaller inputs (optional)

Each of these moves the estimate only a little. Answer any you like, or none.

Wealth that escapes the tax through avoidance and evasion, beyond moving away: which is more credible?

- ○ A. 5%: a small allowance argued from enforcement.
- ○ B. None.
- ○ About equally credible
- ○ Unsure

The billionaires' wealth the tax applies to: which is more credible?

- ○ A. The Forbes list of 1 July 2026: 250 California billionaires worth $2,307 billion, or $2,157 billion without the non-US citizens asked about next.
- ○ B. The Forbes list of 2025: 212 California billionaires, grown 7% to the 31 December 2026 valuation date: $2,027 billion.
- ○ About equally credible
- ○ Unsure

Non-US-citizen residents: which is more credible?

- ○ A. Taxed: the $150 billion of wealth of California billionaires who are not US citizens is in the base.
- ○ B. Not taxed: the wealth of California billionaires who are not US citizens is left out of the base.
- ○ About equally credible
- ○ Unsure

Real-estate deduction: which is more credible?

- ○ A. No deduction: directly held real estate stays in the base.
- ○ B. A deduction: directly held real estate, about 0.5% of the base, is taken out of it.
- ○ About equally credible
- ○ Unsure

Extra income tax from assets sold to pay the tax: which is more credible?

- ○ A. Counted: one third of the tax is paid by selling assets, and the gains on them are taxed.
- ○ B. Not counted: no extra income tax from selling assets is included.
- ○ About equally credible
- ○ Unsure

Why? (optional)

Your answer

PAGE 4 OF 4

Anything else (optional)

Is there an input missing from this list that you think matters?

Your answer

Do you support Proposition 40?

- ○ Yes, I support it
- ○ No, I oppose it
- ○ Undecided
- ○ I prefer not to say

AFTER SUBMIT: THE CONFIRMATION MESSAGE

Thank you for taking the time to answer. All these answers will help us get closer to agreeing on common facts to inform important policy discussions.

See how the two estimates compare: tinyurl.com/prop40-explorer
About the project: tinyurl.com/prop40-opa

Appendix B.2. The treatment form (forms 4 and 6)

Printed from the form's own definition as Google Forms served it in preview on 18 October 2026. Each page of the form starts at a purple rule. Respondents see the text exactly as below, one page at a time.

PAGE 1 OF 5

What are the actual quantities? Proposition 40, input by input

Proposition 40 would tax Californians worth $1 billion or more once, at 5% of their net worth. Estimates of what it is worth to the state depend on a handful of inputs. Each question shows one input and two values that have been proposed for it. Tell us which value you find more credible. Your answers will help us get closer to agreeing on common facts to inform important policy discussions.

Most of these are empirical quantities, and none should depend on whether one supports or opposes a wealth tax.

It takes 5 to 10 minutes. Answers are anonymous unless you leave an email.

* Indicates required question

Do you consider yourself an expert in tax or fiscal policy? *

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Your email (optional)

If you leave one, your answers are also reported in a separate analysis of the respondents who gave an email.

Your answer

How the two estimates differ

Two academic studies estimate what Proposition 40 is worth to California:
• Boll, Saez and Zucman (BSZ, August 2026): +$106.8 billion.
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Two parts explain 83% of the gap: the income tax lost when billionaires leave, and the chance the tax survives in court. Below, how much of the gap each part explains, with each study's values for the first three.

1. The income tax lost when billionaires leave: the share lost, how long it is counted, and billionaires' yearly income tax, together
 $81.3 billion, 56% of the gap
BSZ: 5% of all the income tax that billionaires pay is lost; counted for 5 years; $3.06 billion a year.
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2. Discounting the revenue for court risk
 $39.4 billion, 27% of the gap
BSZ: no discount.
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3. Billionaires who moved out in time
 $29.9 billion, 21% of the gap
BSZ: no billionaire counted by name (its allowance for 5% of the wealth moving away in time is in part 4).
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The questions that follow ask which study's value of each input you find more credible.

The inputs

For each input, pick the value you find more credible.

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Why? (optional)

Your answer

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Why? (optional)

Your answer

The next 5 questions in total explain up to 3% of the difference between the two analyses. Answering them makes the survey on average 55% longer. Answering them is optional. *

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- ○ Unsure

Why? (optional)

Your answer

PAGE 5 OF 5

Anything else (optional)

Is there an input missing from this list that you think matters?

Your answer

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- ○ Yes, I support it
- ○ No, I oppose it
- ○ Undecided
- ○ I prefer not to say

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Thank you for taking the time to answer. All these answers will help us get closer to agreeing on common facts to inform important policy discussions.

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About the project: tinyurl.com/prop40-opa